```
;
; 'Pipe Dream' 22.VI.93 wydruk:
;                              PODSTAWA.MAC
;                              ELEMENTY.MAC
;                              SCR1.LST
;
10        .OPT NO LIST
20 ADRES =   $CB
30 CNT   =   $CD
40 XPOS  =   $D0
50 YPOS  =   $D1
60 HOR   =   10
70 VERT  =   7
80 DELAY =   $19      ; 1/2 sekundy
90 EKRAN =   $4000
0100 SIATKA = $4006
0110 HPOSP0 = $D000
0120 JSTICK0 = $0278
0130 TRIGS = $0284
0140 RANDOM = $D20A
0150 TIMCNT3 = $021C
0160 ;
0170      *=  $3000
0180 ; skoryguj polozenie celownika
0190      LDA #1
0200      STA TIMCNT3
0210 FR   LDA TIMCNT3
0220      BNE FR
0230      STA DLINUM
0240      LDA #$C0
0250      STA $D40E
0260 ;
0270 ; rysuj kratke
0280 KRATA
0290      LDA # <[SIATKA-1]
0300      STA ADRES
0310      LDA # >[SIATKA-1]
0320      STA ADRES+1
0330      STA CLS+2
0340 ; czysc pamiec ekranu
0350      LDY #$0E     ;troche na wyrost
0360      LDA #0
0370      STA CLS+1    ;LO ekranu
0380      TAX
0390 CLS  STA $FFFF,X
0400      DEX
0410      BNE CLS
0420      INC CLS+2    ;HI ekranu +1
0430      DEY
0440      BNE CLS
0450 ;
0460      LDA #VERT+1 ;jedno oczko    wiecej i tak bedzie niewidoczne...
0470      STA CNT     ;licznik poziomow
0480 ;
0490 KR1  LDY #[HOR*2]
0500      LDA #$11     ;pkty w kol. 0101
0510 KR12 STA (ADRES),Y
0520      DEY
0530      BNE KR12
0540      LDA #$01     ;lewe zamkniecie
0550      STA (ADRES),Y
0560 ;
0570      LDX #7
0580 KR2  CLC
0590      LDA #$40
```

```
0600      ADC ADRES
0610      STA ADRES
0620      BCC KR21
0630      INC ADRES+1
0640 KR21 LDY #[HOR*2]
0650      LDA #$01
0660 KR22 STA (ADRES),Y
0670      DEY
0680      DEY
0690      BPL KR22
0700      DEX
0710      BNE KR2
0720 ;
0730      CLC
0740      LDA #$20
0750      ADC ADRES
0760      STA ADRES
0770      BCC CHK
0780      INC ADRES+1
0790 CHK  DEC CNT
0800      BNE KR1       ;nast. oczko
0810 ;    RTS           ;∫·øπ·ß≤∩≈∩«««
0820 ;
0830 ;╟ₘ∩≈∈ßá≡σ∫ₘß
0840 START
0850 ;rzuc hydrant gdzies na ekran...
0860 L1   LDA RANDOM
0870      AND #$07
0880      STA XPOS
0890      INC XPOS      ;xpos= 1-8
0900      JSR SHIFT
0910      LDA RANDOM
0920      AND #$03
0930      STA YPOS
0940      INC YPOS      ;ypos= 1-4
0950      EOR RANDOM
0960      AND #$03      ;4 1.to hydrant
0970      JSR B.O
0980 ;
0990      LDY #$04      ;4 losowania
1000 ;przesun kolejke (Y  razy)
1010 ADDFO
1020      LDX #0
1030 RLL  LDA FIFO+1,X
1040      STA FIFO,X
1050      INX
1060      CPX #3
1070      BCC RLL
1080 ;losuj nowy el. na koncu kolejki
1090 REP  LDA RANDOM
1100      AND #$07
1110      BEQ REP
1120      STA FIFO,X
1130      DEC FIFO,X
1140      DEY
1150      BNE ADDFO
1160 ;wyswietl to w prawym gornym rogu
1170      LDA XPOS
1180      PHA
1190      LDA YPOS
1200      PHA
1210      LDA #HOR
1220      STA XPOS
1230      STX YPOS
```

```
1240 ONE LDX YPOS
1250     LDA FIFO,X
1260     JSR RYSUJ
1270     DEC YPOS
1280     BPL ONE
1290     PLA
1300     STA YPOS
1310     PLA
1320     STA XPOS
1330 TST JSR IFACE
1340     LDY LASTS
1350     BNE TST
1360     INY     .     ;postawiono 1 el                    -uzupelnij kolej.
1370     JMP ADDFO
1380 ;
1390 FIFO .BYTE 0,0,0,0
1400 ;
1410 ;Rysuj element nr 'A(ccum)
1420 ; na poz. XPOS,YPOS
1430 RYSUJ
1440     CLC
1450     ADC #$04    ;pomin hydrant
1460 B.O ;           (bez ograniczen)
1470     ASL A
1480     TAY
1490     LDA IKONKI,Y
1500     STA VEC1+1
1510     STA VEC2+1
1520     LDA IKONKI+1,Y
1530     STA VEC1+2
1540     STA VEC2+2
1550 LICZ.Y
1560     LDA YPOS
1570     ASL A
1580     TAY
1590     LDA RZAD.N,Y
1600     STA ADRES
1610     LDA RZAD.N+1,Y
1620     STA ADRES+1
1630 LICZ.X
1640     LDA XPOS
1650     ASL A
1660     ADC ADRES
1670     STA ADRES
1680     BCC KRESL
1690     INC ADRES+1
1700 ; po "obliczeniach" wyrzuc na ekr.
1710 KRESL
1720     LDY #$00
1730     LDA #$0F    ;wys. 1 elementu
1740     STA CNT
1750     LDX #$00
1760 VEC1 LDA $FFFF,X
1770     STA (ADRES),Y ;na ekran !
1780     INY
1790     INX
1800 VEC2 LDA $FFFF,X
1810     STA (ADRES),Y
1820     DEC CNT
1830     BEQ ON.DSP
1840     TYA
1850     CLC
1860     ADC #$1F    ;$20-1
1870     TAY
```

```
1880      BCC RANGE
1890      INC ADRES+1
1900 RANGE
1910      INX
1920      BNE VEC1
1930 ;juz na ekranie...
1940 ON.DSP
1950      RTS
1960 ; adres wiersza nr 'N (0-6)
1970 RZAD.N
1980      .WORD SIATKA+$20
1990      .WORD SIATKA+$0200
2000      .WORD SIATKA+$03E0
2010      .WORD SIATKA+$05C0
2020      .WORD SIATKA+$07A0
2030      .WORD SIATKA+$0980
2040      .WORD SIATKA+$0B60
2050 ; rurociag...
2060      .INCLUDE #D3:ELEMENTY.MAC
2070 ;        ...i adresy jego  czesci
2080 IKONKI ;         ΣΠ≡ß≤Π≈ßπáδΠΣ·ái
2090      .WORD PG,PD,PP,PL
2100      .WORD DL,DP,GL,GP
2110      .WORD POZIOM,PION,KRZYZ
2120 ;
2130 ;postaw element
2140 FIRE?
2150 ; wyklucz przypadkowe wcisniecie
2160      LDA #DELAY
2170      STA TIMCNT3
2180 PRESS?
2190      LDA TRIG0S
2200      STA LASTS
2210      BNE JOY      ;pomylka...
2220      LDA TIMCNT3
2230      BNE PRESS?
2240 ;
2250      LDA FIFO
2260      JMP RYSUJ
2270 ;
2280 ;
2290 ; Przesuw ramion celownika
2300 LASTJ .BYTE $0F
2310 LASTS .BYTE $01
2320 IFACE
2330      LSR TRIG0S
2340      BCC FIRE?    ;-wcisniety?
2350 JOY LDA JSTICK0
2360      AND #$0F
2370      CMP LASTJ
2380      BEQ IFACE
2390      STA LASTJ
2400 ;to powyzej to chwila wytchnienia
2410 MOVE
2420      LSR A        ;w gore?
2430      BCS NX1      ;...nie
2440      LDA YPOS
2450      BEQ ILLEGAL
2460      DEC YPOS
2470 ILLEGAL
2480      RTS
2490 NX1 LSR A         ;w dol ?
2500      BCS NX2      ;...nie
2510      LDA YPOS
```

```
2520      CMP #VERT-1
2530      BEQ ILLEGAL
2540      INC YPOS
2550      RTS
2560 NX2 LSR A        ;w lewo?
2570      BCS NX3     ;...nie
2580      LDA XPOS
2590      BEQ ILLEGAL
2600      DEC XPOS
2610      BPL SHIFT
2620 NX3 LSR A
2630      BCS ILLEGAL ;joy w spoczynku
2640      LDA XPOS    ;w prawo!
2650      CMP #HOR-1
2660      BEQ ILLEGAL
2670      INC XPOS
2680 ;przesun pionowe ramie
2690 SHIFT LDA XPOS
2700      ASL A
2710      ASL A
2720      ASL A
2730      ADC #$58    ;poz. 1. kratki
2740      STA HPOSP0
2750      RTS
2760 ;
2770 ;ŁŃá·ß≈≤·σá€ßáδΠ€πJi
2780      .INCLUDE #D3:SCR1.LST
;
; To jest plik ELEMENTY.MAC
;
1000 ;------kolanka
1001 DL
1002      .BYTE $00,$00,$00,$01,$00,$00,$00,$01,$80,$00,$EA,$01,$BF,$80,$AA,$C1,$AA,$C0,$AA,$81,
8A
1003      .BYTE $80,$0A,$81,$0A,$80,$2A,$A1,$2A,$A1
1004 ;
1005 DP
1006      .BYTE $00,$00,$00,$01,$00,$00,$00,$01,$00,$02,$02,$AB,$0B,$FE,$0E,$AA,$0E,$AA,$0E,$AA,
0E
1007      .BYTE $82,$0E,$81,$0E,$80,$3A,$A1,$2A,$A1
1008 ;
1009 GL
1010      .BYTE $2A,$A0,$3A,$A1,$0A,$80,$0E,$81,$8E,$80,$BE,$81,$AA,$80,$AA,$81,$AA,$80,$AA,$01,
80
1011      .BYTE $00,$00,$01,$00,$00,$00,$01,$11,$11
1012 ;
1013 GP
1014      .BYTE $2A,$A0,$3A,$A1,$0A,$80,$0A,$81,$0E,$82,$0E,$AB,$0E,$AE,$0E,$AA,$0B,$AA,$02,$AA,
00
1015      .BYTE $02,$00,$01,$00,$00,$00,$01,$11,$11
1016 ;------hydranty
1017 PL
1018      .BYTE $00,$00,$00,$01,$00,$00,$02,$81,$88,$A0,$BE,$21,$AB,$80,$AA,$81,$AA,$80,$AA,$21,
88
1019      .BYTE $A0,$02,$81,$00,$00,$00,$01,$11,$11
1020 ;
1021 PP
1022      .BYTE $00,$00,$00,$01,$00,$00,$0A,$81,$28,$22,$22,$AB,$0B,$FE,$0A,$AA,$0A,$AA,$22,$AA,
28
1023      .BYTE $22,$0A,$81,$00,$00,$00,$01,$11,$11
1024 ;
1025 PD
1026      .BYTE $00,$00,$00,$01,$00,$00,$0A,$81,$28,$A0,$22,$21,$0B,$80,$0E,$81,$0E,$80,$2E,$A1,
0E
```

```
1027      .BYTE $80,$0E,$81,$0E,$80,$3A,$A1,$2A,$A1
1028 ;
1029 PG
1030      .BYTE $2A,$A0,$3A,$A1,$0A,$80,$0A,$81,$0E,$80,$2E,$A1,$0E,$80,$0E,$81,$0B,$80,$22,$21
28
1031      .BYTE $A0,$0A,$81,$00,$00,$00,$01,$11,$11
1032 ;------rury
1033 KRZYZ
1034      .BYTE $2A,$A0,$3A,$A1,$0A,$80,$0A,$81,$8E,$82,$EE,$AB,$AE,$BE,$AE,$AA,$AE,$AA,$AE,$AA
8E
1035      .BYTE $82,$0E,$81,$0E,$80,$3A,$A1,$2A,$A1
1036 ;
1037 PION
1038      .BYTE $2A,$A0,$3A,$A1,$0A,$80,$0A,$81,$0E,$80,$0E,$81,$0E,$80,$0E,$81,$0E,$80,$0E,$81
0E
1039      .BYTE $80,$0E,$81,$0E,$80,$3A,$A1,$2A,$A1
1040 ;
1041 POZIOM
1042      .BYTE $00,$00,$00,$01,$00,$00,$00,$01,$80,$02,$EA,$AB,$AF,$FE,$AA,$AA,$AA,$AA,$AA,$AA
80
1043      .BYTE $02,$00,$01,$00,$00,$00,$01,$11,$11
;
; ...a to SCR1.LST
;
01 POINT =   4        ;kolor celownika
02 BKGRD =   2        ;kolor tla
05 COLPM0 =  $D012
09 ;obsluga DLI - celownik na ekranie
10 DLINT
20        PHA
30        TXA
40        PHA
50        LDA #POINT  ;postaw pasek
60        LDX DLINUM
70        BNE NIE0
80        STA COLPM0  ;...pionowy
90 NIE0   CPX YPOS
0100      BEQ ROWNE
0110      LDA #BKGRD  ;usun pasek
0120 ROWNE
0130      STA COLPM0+1 ;...poziomy
0140      STA COLPM0+2
0150 ;
0160      INC DLINUM
0170      CPX #VERT
0180      BCC EXITDLI
0190      STA COLPM0  ;...i  pionowy
0200      LDA #0
0210      STA DLINUM
0220 EXITDLI
0230      PLA
0240      TAX
0250      PLA
0260      RTI
0265 DLINUM .BYTE $00
0270 ;
0280 ; DL'ka  z DLI
0290 DLIST
0300      .BYTE $70,$70,$70,$70,$70,$70
0310      .BYTE $CE, <EKRAN, >EKRAN
0320      .BYTE $0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$8E
0330      .BYTE $0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$8E
0340      .BYTE $0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$8E
0350      .BYTE $0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$8E
```

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0360      .BYTE $0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$8E
0370      .BYTE $0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$8E
0380      .BYTE $0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$0E,$8E
0390 ; okienko dla edytora
0400 ;    .BYTE $42,$40,$BC,$42,$68,$BC,$42,$90,$BC,$42,$B8,$BC,$42,$E0,$BC,$42,$08,$BD,$42,$30
BD
0410      .BYTE $41, <DLIST, >DLIST
0420 ; zezwol tylko na VBLK'e
0430      *=  $D40E
0440      .BYTE $40
0454 ; wektor DLI
0455      *=  $0200
0456      .WORD DLINT
0460 ;dane dla ANTIC'a
0470      *=  $022F
0485      .BYTE $21, <DLIST, >DLIST
0486 ;priorytet wskaznika
0487      *=  $026F
0488      .BYTE $04
0490 ;kolory
0500      *=  $02C0
0510      .BYTE BKGRD,BKGRD,BKGRD,BKGRD,$00,$08,$0E,BKGRD,BKGRD
0550 ;pozycja poziomego paska
0560      *=  $D001
0570      .BYTE $58,$78,0,0,$98,$A0,0
0580 ;dane wskaznika
0600      .BYTE 0,3,3,0,$3C
0610      .BYTE $FF,$FF,$FF,0,$3C
```